

KAZ POWER PLANT UPGRADE PROJECT

JOB HAZARD ASSESSMENT (JHA) - REVISED

SE GT SU | Job-based hazard analysis is performed in A, B, C, Small projects and, by decision of EHS managers, in all projects.

Activity	Kaz Power Plant Upgrade Project (Construction, Installation and Testing of SS Extension)
Location	Basrah, Iraq
Work Area	Khur Alzubair Power Station
Document No.	1 (Original Date: 04.03.2026 - Revised: 13.09.2026)
Team Member	Ugur Akbulut, Onur Ozvatan, Abdurrahman Ogut, Dogukan Arandi
Used Vehicles	Manlift, Crane, Truck, Excavator
Used Equipments	Hand Tools, Power Tools, Lifting Equipments, Test Equipments
Persons Affected	Operators, maintenance, visitors, public, etc.
Review	If there is a change in working methods and conditions, the form must be renewed.

POLICE	112	FIRE STATION	112	AMBULANCE	112
SIEMENS ENERGY EMERGENCY CONTACT NUMBER	+90 533 625 9494				

NOTE: If after risk evaluation and with all the necessary controls in place, the risk rating (R) is HIGH, the EHS department are to be contacted for guidance on the task/process.

Step 1	The existing general potential risks and hazards in the work area are defined. The relevant risk is ticked. 26 risk groups were defined (page 2).
Step 2	Existing work steps are written one by one and work equipment is specified on a separate line for construction equipment and subjected to hazard assessment. Severity and likelihood are entered for each step using the 5x5 Siemens matrix; R = S x L is entered as Low / Medium / High. Control measures are evaluated and the risk score decreases in line with the measures taken / to be taken (pages 3-9).
Step 3	Relevant PPE is selected to protect against the risks mentioned in Step 2; under each PPE the related risk number from Step 2 is referenced to confirm (page 10).
Step 4	Approval section. The site manager / field supervisor and the OHS specialist conduct the final review and mark SUITABLE / UNSUITABLE. If 'Not Suitable' is indicated, this hazard assessment must be redone - work cannot start (page 11).
Step 5	The approved JHA is communicated to employees; employees acknowledge by writing their name, surname and signature. Work starts after Step 5 is completed (page 11).

No.	Risk Group	Work Permits Related to the Risk Group (Step 3)	Ticked
1	Electric shock or exposure to electric arc	ELECTRICAL WORK PERMIT	X
2	Fall from height	WORKING AT HEIGHT PERMIT	X
3	Falling object from height	WORKING AT HEIGHT PERMIT	X
4	Hot work (oxy-LPG, oxy-acetylene, LPG, torch)	HOT WORK PERMIT	
5	Welding (electrode, aluminum)	HOT WORK PERMIT	X
6	Ergonomic strain / manual handling	JHA	X
7	Hazardous waste	MOTAT Records	X
8	Use of mobile electrical panel and extension cable	ELECTRICAL WORK PERMIT	X
9	Use of crane, hoist, forklift, compressor, personal carrier, etc.	CRANE WORK PERMIT, LIFTING PLAN	X
10	Use of construction machinery (excavator, truck)	EXCAVATION WORK PERMIT	X
11	Angle grinder use	JHA	X
12	Use of chemical substances (ethyl alcohol, acid, base, epoxy, etc.)	JHA	X
13	Waste generation due to activity (cable, metal, paper, hydrocarbon, etc.)	MOTAT Records	X
14	Explosive areas	JHA	
15	Exposure to dust, vapor, gas, welding fumes, chemical and biological agents	JHA	
16	Moving machine parts or work equipment (entrapment, impact, etc.)	JHA	X
17	Slip and trip hazards on the ground	JHA	X
18	Contact with extremely cold or hot surfaces	JHA	
19	Working near energized open busbars and conductors in low-voltage systems	ELECTRICAL WORK PERMIT	X
20	Working in high-voltage system (≥ 1000 V)	ELECTRICAL WORK PERMIT, EKAD	X
21	Environmental factors, working in extremely hot & cold climate	JHA	
22	Hand tool use (hammer, cutting tools, etc.)	JHA	X
23	Use of scaffolding, ladder and platform	WORKING AT HEIGHT PERMIT	X
24	Insufficient lighting and ventilation	CONFINED SPACE WORK PERMIT	X
25	Working in confined or restricted space	CONFINED SPACE WORK PERMIT	X
26	Use of pressure vessel	CONFINED SPACE WORK PERMIT	

SIEMENS 5x5 MATRIX: R = S x L | 1-6 LOW (Acceptable, Monitor & Review) | 8-12 MEDIUM (Acceptable With Controls) | >12 HIGH (Unacceptable) | 1. Identify risks/hazards WITHOUT control. 2. Record controls and repeat the assessment with controls in place. | NOTE: Commissioning activities removed in this revision.

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
1	BEFORE WORK	Unauthorised / untrained persons entering the site Absence of valid work permit before mobilisation Missing or expired qualification documents for skilled trades Personnel not medically fit for the task Exposure to multiple simultaneous site hazards (live electrical equipment, lifting operations, heavy machinery) Extreme ambient (hot/cold) working conditions	P.A. E	Personal injuries Property / asset damage Environmental impact Regulatory non-compliance	4	3	12	M	1- Ensure a valid work visa and valid employment documents are available before going to the country for work. 2- Strict compliance to PTW system - the safety work permit and all pertinent documents shall be secured before work commence. 3- Customer safety induction training and SE EHS field induction training to be taken before starting work at the construction site; the training attendance report must be signed by each employee. 4- Siemens Energy and the customer risk assessment must be properly read, understood and discussed with all personnel before starting work. 5- A competent person must be appointed for the specific task. For certain jobs (crane operator, forklift operator, welder, scaffold builder, electrician, mechanical maintenance man, etc.) the necessary vocational training and qualification documents should be requested. Unauthorized personnel should not work outside of the job description. 6- A health report approved by the doctor for the last year (or valid in the home country) must be available for all personnel. 7- Minimum PPE to be provided and worn by each employee: Helmet EN 397 (chin-attached), work shoes EN 345 S3 with ankle protection, safety glasses EN 166, ear protection, protective work gloves EN 388, protective dust mask (FFP3), high-visibility clothing, disposable overalls, head lamp. 8- All crew must be familiar with the emergency management plan and actions from the customer (nearest hospital, response timing, muster points). 9- Keep your passport and some local cash in case of unexpected events (earthquake, fire, explosion, etc.). 10- Daily working hours cannot exceed 11 hours during the day and 7.5 hours at night, excluding rest periods. 11- Seven days of uninterrupted work cannot be done; one day of rest is obligatory within seven days. Annual overtime cannot exceed 270 hours. 12- Certified First Aider with valid certificates must be present at site; first aid kit to be stocked and accessible. 13- Emergency procedure to be followed. Stop Work Authority applies at all times. NO HSE - NO WORK.	4	1	4	L
2	GENERAL SITE WORK	Uncontrolled site access and activity Unreported site risks / unsafe conditions observed during work Work performed without valid permit Inadequate supervision and communication	P.A	Personal injuries Property loss Work stoppage Regulatory non-compliance	4	3	12	M	1- Fill the "LMRA" form before starting work and get the necessary work permit from the client; strict compliance to PTW requirements. 2- Conduct TBT at the start of each shift; the RA to be discussed daily with the workers at the start of work. 3- When risks that may affect personnel are observed at the project site, inform the customer immediately and ask for action. 4- If the client does not take any action, notify the project manager. 5- Assign competent supervision at all times; proper signages at site. 6- Barricade and segregate the work areas before starting the activity. 7- Minimum PPE to be worn at all times: Helmet EN 397, work shoes EN 345 S3, safety glasses EN 166, gloves EN 388, high-visibility vest. 8- Emergency procedure to be followed. Stop Work Authority applies. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
3	DRIVING - TRANSPORT TO SITE	Road traffic accident (collision / overturning) Speeding and disobedience to local traffic regulations Driver fatigue Mobile phone / device use while driving Unsecured or oversized loads Adverse weather conditions (rain, fog, icing) Wild or domestic animals on the road Defective vehicle / tyres not suitable for season	P.A. E	Serious injury, death, asset damage and 3rd party consequences (if happen on the road), fire	3	3	9	M	1- Everyone must have a valid driver's license for the type of vehicle driven and be medically fit to drive, in accordance with local regulations. 2- Vehicle must be approved from third party and client; daily vehicle inspections by driver before departure. 3- Trained personnel, certified operator only. 4- Obey all local traffic regulations and speed limits while driving. 5- Seatbelt compliance at all times. 6- Plan journeys taking into account distance, road conditions, travel time and weather conditions. 7- No one should drive while under the influence of drugs or alcohol. 8- No one should use mobile phones or other devices that cause distraction while driving, including in hands-free mode - this is unsafe and unacceptable. Minimize distraction (eating, drinking, radio/GPS setting). 9- No one should drive when tired or suffering from fatigue; take adequate breaks at least every two hours. If you feel tired, stop and rest. 10- Ensure the loads carried are within the vehicle weight and load limits and are properly secured before departure. 11- Be careful about the possibility of wild or domestic animals on the road. 12- Use the vehicle considering weather conditions (rain, snow, fog, icing); tyres suitable for seasonal conditions. 13- Road survey for obstacles and overhead structures; flagman or escort vehicle if required. 14- Vehicle must be equipped with fire extinguisher and first aid kit; certified first aider must be available. 15- Emergency procedure to follow. Stop Work Authority applies. NO HSE - NO WORK.	3	1	3	L
4	GENERAL SITE WORKS (STOP! - STOP WORK AUTHORITY)	Unsafe acts / unsafe conditions left uncorrected Non-compliance with Zero Harm behaviours, essentials, local programs or legal requirements Potential harm due to overconfidence / normalization of risk Failure to stop work in high-risk situations	P.A	Injury, fatality, asset damage, legal consequences	4	4	16	H	1- Everyone is empowered to apply STOP! any time there is a potential risk of harm or non-compliance with Zero Harm Behaviours, Essentials, local Zero Harm programs, or legal requirements. 2- Taking into consideration the severity of the circumstances, STOP! may be issued by anyone as follows: - Immediate direct verbal STOP!, usually involving minor non-compliance, including unsafe acts or conditions that can be easily corrected on the spot without delay. - Immediate notification of STOP! to the responsible person in charge of the work activity or workplace condition for immediate corrective action. 3- Everyone experiencing a STOP! notification from another individual is expected to receive it in a constructive way and as a kind reminder of the Zero Harm Principle "We Take Care of Each Other". 4- Everyone must respect the STOP!, and it shall remain in force until the required corrective measures have been put in place. 5- When in doubt how to resolve a situation leading to a STOP!, everyone must consult their managers and verify the applicability and implementation of required measures. 6- NO HSE - NO WORK. Emergency procedure to follow.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
5	INCIDENT MANAGEMENT	Unreported incidents / near misses Delayed reporting of injuries (even minor ones) Failure to learn lessons from incidents Inadequate corrective / preventive actions Re-occurrence of similar events	P.A	Personal injuries Recurrence of incidents Property loss Reputational damage	4	3	12	M	1- Everyone needs to know the Siemens Energy definitions and procedures for reporting incidents and near misses. 2- Any incident or near miss that a person is involved in or witnesses must be reported to the manager or supervisor without delay. 3- Anyone who experiences an injury, no matter how minor, or sudden pain from a work activity must speak up without delay and report the situation to the manager or supervisor. 4- Anyone involved in or witnessing an incident / near miss will support the investigation process and contribute to cause analysis, lessons learned and closure of the incident. 5- Everyone should be familiar with the corrective and preventive measures required following the investigation process and be fully aware of any changes in procedures or risk assessments covering business activities. 6- Support the implementation of corrective and preventive measures by proactively checking that everyone you work with is fully aware of changes. 7- Daily TBT to reinforce incident reporting culture; emergency procedure to follow.	4	1	4	L
6	WORKING UNDER ELEVATED PHYSICAL STRESS AND MANUAL LIFTING / HANDLING	Manual handling of heavy / awkward loads Work that is too frequent or too long, requires extreme distances, or is done with insufficient rest Incorrect lifting / carrying posture Load characteristics (size, weight, manageability, center of gravity, external/internal hazards) Insufficient space, uneven ground and adverse climatic conditions in the study area	P	Musculoskeletal injuries, strains, back injuries, fractures	3	3	9	M	1- All personnel to be familiar with correct handling / lifting / carrying techniques (lift using the legs and arms, not the back). 2- Seek assistance when attempting to move heavy loads; use auxiliary equipment for lifting and transporting where possible. 3- Plan for adequate breaks; avoid work that is too frequent or too long, requires extreme distances, or is done with insufficient rest time. 4- Take into account the physiological state of the employee; assign only those personnel who are physically fit to perform the task. 5- The characteristics of the load (size, weight, manageability, location of the center of gravity, external and/or internal hazards) shall be considered before handling. 6- Features of the study area (insufficient space, uneven ground, climatic conditions) shall be taken into account in the handling method. 7- Provide appropriate instruction on the correct handling of loads and the hazards arising from misuse. 8- Proper PPE: Helmet EN 397, work shoes EN 345 S3, gloves EN 388. 9- Conduct TBT on manual handling; emergency procedure to follow. NO HSE - NO WORK.	3	1	3	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
7	WORKING AT HEIGHT	Fall of persons from height Falling of objects / tools from height Defective, unapproved or unlabelled scaffolding (red label in use) Unguarded edges, openings and platforms Excessive wind speed Working at height without fall protection equipment	P.A	Fatality Serious personal injury Property damage	5	3	15	H	1- Scaffolding installation / dismantling must be done by competent and authorized personnel; the necessary documents regarding scaffolding suitability will be submitted by the customer to Siemens Energy before the work. 2- Scaffolding should only be installed or replaced by a specialized company; all scaffolding to be inspected weekly by a trained and qualified scaffolding inspector. 3- Only access inspected and approved scaffolding - green label is suitable, red label is not suitable. ScaffoldTag to be displayed. 4- Every employee should check the scaffolding to observe obvious deficiencies before going on the scaffolding. 5- Personnel working on the scaffold must use the necessary PPE: full body harness EN 361 with double lanyard (yoyo) and lifeline - 100% tie-off. 6- In the work to be done on the GT, the use of yoyos attached to the lifeline should be a priority. In cases where lifelines need to be dismantled due to the requirements of the lifting operation, the yoyos should be connected to the ceiling crane, anchor points should be used, or the yoyos and the turbine should be anchored at appropriate points to be determined and seat belts should be used. 7- Materials and equipment should not be left on scaffolding, access platforms, etc. at the end of the work or during shift change; the risk of objects falling should be eliminated. 8- Work should not be done on scaffolds where guardrails and barriers have not been installed; all openings and edges to be closed. 9- Each study should be evaluated on its own and collective protection measures should be given priority; where collective protection measures cannot be taken, precautions should be taken with personal protective equipment and equipment. 10- Minimum PPE for work at height: Helmet EN 397 with chin strap, full body harness EN 361 with double lanyard, work shoes EN 345 S3, safety glasses EN 166, gloves EN 388. 11- Conduct TBT before start of work; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	5	1	5	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
8	WORKING WITH BUILDING / CEILING CRANE	Dropped load / swinging load Crane failure or defect Untrained / uncertified crane operator Defective lifting gear / load rigging with expired inspection Unclear or multiple signals to the crane operator Working under a suspended load Load lifted above work areas with personnel below Chain snagging during lifting with the crane hook	P.A	Personal injury, fatality, property / asset damage	4	3	12	M	1- Only employees with an authorized overhead crane operator certificate can use the overhead crane; only appropriately trained and qualified personnel to be used. 2- Periodic inspection reports should be provided before the overhead crane is used; before the lifting operations, the visual control of the lifting equipment should be done and periodic control reports should be provided. 3- Prepare a lifting plan if complex crane operations are to be performed; plan all crane activities in advance and prepare the set-down area. 4- Lifting gear / load rigging shall be visually inspected before use; notify supervisor of defective lifting gear / load rigging; mark the defective lifting gear red and block for further use. 5- Do a walkdown of the lift path before crane work and warn personnel working in the hazardous area. 6- Ensure clear and precise signals between rigger and crane operator; only the rigger shall give signals to the crane operator. 7- Do not give any direction commands while the weight is swinging from the crane hook; wait until the weight has come to a standstill. 8- When lifting with the crane hook, take care to ensure that the chain does not become snagged. 9- NEVER work under a suspended load; never turn your back on the load. 10- All personnel shall clear the hazardous area if the load is to be moved above work areas; keep personnel who are not required out of the hazardous area. 11- Use chain hoist for small lifting operations or use chain hoist on crane hook; use the actual crane for larger operations. 12- Use edge protection to prevent damage to lifting gear due to sharp edges; maintain distance from load; keep transport routes free. 13- Always use the correct device for lifting operations; consult your supervisor if you are uncertain. 14- PPE to be worn: Helmet EN 397, work shoes EN 345 S3, gloves EN 388, safety glasses EN 166, high-visibility vest. 15- Conduct TBT before start of work; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
9	HAZARD DUE TO ELECTRICAL CURRENT, HIGH VOLTAGE	Electric shock, electrocution and electric arc flash exposure Unexpected energization of equipment Working near energized open busbars and conductors in low-voltage systems Working in high-voltage systems (>=1000 V) Uncontrolled release of electrical energy Inadequate isolation / LOTO application Use of unshielded electrical equipment in confined areas	P.A. E	Electric shock, burns, fatality, equipment damage, fire, explosion	4	4	16	H	1- Strict compliance to PTW - the electrical work permit must be taken before any electrical work; safety instructions must be followed. 2- Compliance with the 5 safety rules (to be applied by the customer during electrical work): 1- Isolate 2- Lock Out / Tag Out 3- Confirm that equipment cannot be re-energized 4- Grounding and short circuit 5- Maintain isolation with adjacent energized elements 3- Work only under a valid permit to work prepared and issued by a person tested and authorized by the customer. 4- Use only the tools appropriate for the activity; use shielded extra low voltage (PELV) or safety extra low voltage (SELV) electrical equipment in confined areas. 5- Only electrically trained and certified employees should work in electrical work; assign the task to a competent person. 6- Wear appropriate PPE: Helmet EN 397, arc-rated protective clothing, insulated safety gloves, face shield / safety glasses EN 166; insulating mats to be laid in channels where energized MV cables are located. 7- Barricade & keep safe distance from live areas; clear identification of live equipment; maintain safe clearance. 8- Verify line status and pressure before work; energy isolation where possible with approved isolation certificate and valid PTW. 9- Stop work immediately if safe distances cannot be maintained. 10- Emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
10	WORKING WITH FORKLIFT AND MOBILE CRANES	Vehicle struck / collision / overturning Untrained or unauthorized operators Defective equipment, missing periodic inspection Loads falling during lifting / transport Lifting or conveying personnel with non-compliant equipment Working inside the crane / forklift maneuvering area Blind spots around vehicles	P.A	Serious injury, death, asset damage	4	3	12	M	1- Before using forklifts and mobile cranes, obtain periodic control documents, periodic maintenance records, and the operator certificates of the employees who will use them. 2- Third party certified operator shall operate the forklift / boom truck; the driver must have a valid driving license as per law. 3- Except for the relevant operators, forklifts and mobile cranes should not be operated by other employees. 4- Daily inspection checklist to be completed by the operator before use; any defect to be reported immediately. 5- For forklifts, reverse gear warning alarm and strobe warning light will be in working condition; audible reverse alarm is mandatory. 6- No work should be done in the mobile crane and forklift operation and maneuvering area, and personnel should be removed; a flagman with reflective vest to be assigned. 7- Lifting plans should be considered in lifting operations; forklift transports should be carried out considering the center of gravity of the load. 8- Lifting and conveying personnel should not be done with mobile cranes, overhead cranes and forklifts without equipment complying with the standards. 9- In order to prevent the parts from falling during lifting, the parts should be lifted with durable and closed systems such as boxes and crates. 10- Vehicle must be equipped with fire extinguisher and first aid kit. 11- Emergency procedure to follow; strict compliance to PTW. NO HSE - NO WORK.	2	3	6	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
11	HOT WORK (e.g., grinding, welding, cutting)	Fire / explosion from sparks, hot work and uncontrolled heat sources Welding fumes and harmful gas exposure Compressed gas cylinders - flashback, valve damage, cylinder projection / explosion Defective or untested gas hoses and connections Portable electrical tools (faulty cords, un-earthed equipment) Pinch points and rotating equipment Combustible materials present in the working area	P.A. E	Fire, burns, serious injury, death, asset damage, health effects (chronic illnesses, cancer)	4	3	12	M	1- Work permit must be taken before work; strict compliance to PTW and JHA requirements. 2- Mark areas and set up barricades if necessary; remove flammable materials from the environment (maintain a 11-meter clearing radius around all welding and cutting areas, or provide suitable fire-resistant shielding where removal is not practicable). 3- Have an observer / fire watchman available in case of fire; conduct post-welding fire watch (30 min). 4- Keep fire fighting equipment with you; fire extinguishers to be available at all times and operated by a trained person. 5- Wear appropriate PPE: Helmet EN 397, work shoes EN 345 S3, protective work gloves EN 388, ear protection; use full-coverage goggles or face visor for cutting, grinding, etc. 6- Ensure good ventilation in the area (natural or forced); MSDS of welding electrodes to be available. 7- All oxygen and propane / LPG hoses to be equipped with flashback arrestors; cylinders to be fixed in trolleys or chained to the wall in vertical position to prevent falling; when not in use the valve cups must be placed back. 8- Liquid gas cylinders (propane or butane) to be stored at least 3 meters from ALL other types of gas cylinders; NO SMOKING in the oil storage area. 9- Use of fire blankets to prevent sparks flying outside the working area; good housekeeping to avoid accumulation of combustible waste. 10- Tools to be inspected daily by the user; experienced / trained workers only; TBT on correct use of portable tools; continuous supervision. 11- Certified first aider must be available at site; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
12	FIELD ACTIVITIES (Trips / Falls / Injuries)	Materials falling or tipping from unorganized stacking Material transport with unsuitable or overloaded vehicles Defective slings used for material transport Working under a load being lowered by crane First aid / fire equipment not available in vehicles Adverse weather during field activities (rain, fog, heavy weather)	P.A	Personal injuries, property loss, fatality	4	3	12	M	1- Materials being transported must be stacked in an organized manner; materials should be stacked in separate areas according to their size and weight, ensuring they do not tip over or collapse. 2- Materials should be transported using appropriate vehicles; first aid kits and fire extinguishers must be available in the vehicles. 3- Slings used for material transport must undergo visual inspection; worn slings should not be used. 4- Do not stand under the load while lowering materials with a crane; do not lift loads exceeding the crane's capacity. 5- Material transport and service vehicles should not be operated by anyone other than authorized and trained personnel. 6- Work should not be carried out in rainy, foggy, or heavy weather; weather reports should be checked before work begins. 7- Barricade and signage at material handling areas; keep transport routes free. 8- Minimum PPE: Helmet EN 397, work shoes EN 345 S3, safety glasses EN 166, gloves EN 388, high-visibility vest. 9- Conduct TBT before start of work; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
13	PPE & EQUIPMENT	Absence of required PPE for the activity Defective, damaged or expired PPE PPE not suitable for the task / exposure Inadequate PPE allocation, record-keeping and periodic inspection	P	Personal injuries of all exposure types, fatality	4	3	12	M	1- At the start of work, all personnel's PPE assignment forms are checked using the personnel on-the-job control form; PPE deliveries should be made in the form of notification and receipt. 2- PPE should be checked at least once a month and the checks should be recorded; if employees detect any malfunction or damage to their personal protective equipment, they should inform their supervisor and ensure it is replaced. 3- Minimum PPE: 1- Helmet EN 397 (chin strap for teams working at height) 2- Work shoes EN 345 S3 or S1P (with sole and toe protection), safety glasses EN 166 (anti-fog), gloves EN 388 (for employees); EN 361 safety harness should be used for work at height. 3- In addition, appropriate personal protective equipment (safety harnesses, welding masks / gloves, dust masks, etc.) according to the nature of the work should be delivered and recorded on the PPE assignment form. 4- Company logo work outfit should be worn. 5- Supervisor / Foreman has to make sure all crew members have the required PPE as per activity before the start of work. 6- NO HSE - NO WORK. Emergency procedure to follow.	4	1	4	L
14	INSTALLATION OF SWITCHYARD EQUIPMENT	Dropped load during lifting of switchyard equipment Crane failure / overturning on soft or uneven ground Slinger / signalman not qualified or not directing the operation Simultaneous top-bottom (stacked) work during assembly Pylon climbing without vertical lifeline / fall protection Personnel entering under the suspended load Adverse weather (high wind) during lifting	P.A	Serious injury, fatality, asset damage	4	3	12	M	1- Before work begins, safety approach / jump distances must be evaluated; the power should be cut off at points within the safety distance, and the 5 safety rules should be applied. 2- Devices should be lifted using a mobile crane that has undergone periodic checks and has the appropriate and inspected legal documents; a crane work permit must be filled out. 3- There must be a slinger connecting the devices and a signalman directing the operation; only trained and qualified personnel to be used. 4- The crane legs must be fully extended to the solid ground; in case the legs sink, materials that will expand the surface area (pads) should be placed under the legs. 5- Material guiding ropes (tag lines) should be used; the operation should be carried out taking weather conditions into consideration (operations suspended for winds exceeding 50 km/h). 6- No personnel should enter under the material under any circumstances; the work area should be restricted, barricaded and fenced off with warning signs; no one other than the signalman / slinger should be in the area; the slinger / signaler should under no circumstances go under the lifted load. 7- For all work to be done with the crane, the "CRANE WORK CHECKLIST" must be filled out and approved. 8- During assembly, it should be checked whether simultaneous (top-bottom) work is being carried out; any simultaneous work should be stopped immediately. 9- Slingers guiding during assembly should not be under the pylon / beam / load. 10- Climbing the pylon should be done using a vertical lifeline and/or yo-yo; walking on the equipment should be done using a horizontal lifeline; full body harness EN 361 with double lanyard to be used. 11- Minimum PPE: Helmet EN 397 with chin strap, EN 361 harness, work shoes EN 345 S3, gloves EN 388. 12- Conduct TBT before start of work; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
15	WORKING IN CONFINED SPACES (inadequate evacuation and rescue possibilities, poor communication to the outside, risk of panic attacks, poor ventilation conditions, insufficient room to move)	Hazardous / explosive atmospheres inside the space Oxygen deficiency / asphyxiation Working alone without a watchman Untested atmosphere before entry Poor communication with the outside Intrinsically unsafe electrical equipment used inside the space	P.A	Suffocation, asphyxiation, intoxication, explosion, multiple fatalities, asset damage	4	3	12	M	1- Confined space work permit must be obtained before work; strict adherence to PTW system. 2- Perform a visual inspection of tagging and locking for correct application before entering confined spaces. 3- Ambient air measurement (oxygen concentration, gas testing) should be performed before entering the enclosed space and at regular intervals; authorized gas testing only. 4- Assign a watchman / CS attendant at the entrance manhole and maintain constant contact with the watchman; rescue equipment to be kept ready if necessary. 5- Use only safe extra low voltage (SELV) or shielded extra low voltage (PELV) / intrinsically safe (Ex) electrical equipment inside the confined space. 6- Maintain order and cleanliness (proper housekeeping) in confined spaces during maintenance work. 7- It is forbidden to work alone; must have at least 2 employees; appropriate persons should be assigned to the job. 8- Respirator should be used if necessary; ventilation equipment must be available. 9- When accessing the pier, only the access point provided for this purpose should be used; access and exit points should be clearly defined; emergency exits must be kept open at all times. 10- Hard barriers and warning signs at the entrance; first aider at site. 11- Emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
16	PORTABLE LADDER USAGE	Fall from defective / damaged ladder (broken, bent or deformed legs) Ladder not suitable for the work height Insufficient securing / anchoring of the ladder Use of the top two steps as a working position Working with both hands occupied (e.g., stone cutter) for a prolonged time Ladder overloaded beyond rated capacity Incorrect ladder angle on unstable surface	P	Fall injury, fractures, fatality	4	3	12	M	1- Ladders must be selected according to the height at which the work will be performed; ensure the ladder is long enough to safely reach the work area. 2- Ladders must have a CE Certificate with manufacturer information and a label indicating CE standards, and must comply with the EN 131 standard. 3- Work should not exceed 30 minutes on ladders, and heavy work requiring prolonged use of equipment with both hands (cutting with a stone cutter, etc.) should not be performed. 4- Ladders should be checked before each use; inspection cards should be available and checked at least once a week. 5- On type A ladders, the securing strap or locking mechanism must be completely locked when the ladder is opened; all locks should be properly engaged and the two sections to overlap according to the manufacturer's instructions. 6- Ladders with deformed, broken, or bent legs should not be used for work; ensure the ladder is in good condition, not loaded beyond the maximum intended load, placed against a stable surface at the correct angle (4:1 ratio - 75 degrees). 7- Contact should be made at least at 3 points on the ladder during work; work should not be done on the top two steps of the ladder. 8- The ladder to be used for climbing must be sufficiently strong and secured by securing it at a suitable point; in cases where securing is not reasonable or possible, ladder securing should be done according to the manufacturer's instructions. 9- PPE: Helmet EN 397 with chin strap, work shoes EN 345 S3, gloves EN 388. 10- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
17	WORKING WITH HAND TOOLS / POWER TOOLS / MEASURING EQUIPMENTS	Use of tools not appropriate for the activity / defective tools (missing inspection tag) Flying particles and slung parts (cuts, eye injury) Hand / arm vibration from power tools Entanglement in rotating parts Untrained / unauthorized personnel operating equipment Unsafe routing of electrical / pneumatic / compressed air lines Hydraulic / pneumatic whip lash and fluid injection Work performed without following the safety instructions in the manual	P.A	Cuts, lacerations, fractures, eye injury, hearing damage, property damage	4	3	12	M	1- Only use tools which are appropriate or intended for the activity - right tool for the right job (never use a file as a lever or a screwdriver as a chisel). 2- Visual inspection before use; check the inspection tag and color code; check to ensure that safety equipment is intact; defective tools to be removed from the worksite immediately. 3- Safety instructions in the instruction manual shall be observed. 4- Wear appropriate PPE: safety glasses EN 166 / face shield, gloves EN 388, ear plugs / muffs for noisy operations, work shoes EN 345 S3, Helmet EN 397. 5- Use low-vibration equipment or schedule sufficient breaks. 6- Secure tools in such a way that slipping away during operation is prevented; shield adjacent work areas if injuries to others are possible due to flying particles or slung parts. 7- Use only appropriately trained and qualified personnel. 8- Lines (electrical, pneumatic, compressed and process air) shall be safely routed and secured. 9- If hydraulic fluid is used, follow the instructions in the safety data sheet or the operating instructions; use whip check when using hydraulic and pneumatic systems. 10- Conduct TBT before start of work; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
18	HAZARD DUE TO FALLING (objects from height)	Objects / tools falling from scaffolding, platforms or equipment Loose materials left at height at the end of a shift Unguarded edges, openings and platforms Personnel working or walking below the drop zone without protection	P.A	Struck-by injuries, fatality, property damage	4	3	12	M	1- Installation of scaffolding or barriers; only access scaffolding that has been inspected and approved (green label). 2- Observe the work instruction on use of PPE to prevent falling; use PPE to prevent falling (full body harness EN 361 with 100% tie-off where required). 3- Use tool lanyards / keepers, toe boards and mesh under platforms to prevent falling of objects and tools. 4- Barricade the area below the work at height and keep personnel out of the drop zone; warning signs to be placed. 5- No materials or tools to be left unattended on platforms at the end of the work or during shift change. 6- Minimum PPE: Helmet EN 397 with chin strap, work shoes EN 345 S3, gloves EN 388. 7- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
19	HAZARD DUE TO SLIPPING, STUMBLING AND FALLING ON SAME LEVEL	Wet / oily / lubricated surfaces Debris, waste and material accumulation in walkways Trailing cables and hoses Uneven floor, gaps or deformation Poor lighting in walk areas	P.E	Minor and major fall injuries, fractures	3	3	9	M	1- Lay cables outside the hazard zone; all cables and hoses to be neatly routed and secured to prevent slips and trips. 2- Materials and tools which are not required shall be removed from the work area; maintain housekeeping standards at all times. 3- Traffic routes shall be kept clear; walkway and gangway to be kept free of obstruction. 4- Make sure that there is no deformation, lubrication or gap on the work area floor, and take the necessary precautions if any (non-slip mats, drainage); remove water / oil spills immediately. 5- Adequate lighting in work areas; proper footwear (work shoes EN 345 S3 with anti-slip sole) to be worn. 6- Signage to be placed where temporary hazards exist. 7- Emergency procedure to follow. NO HSE - NO WORK.	3	1	3	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
20	HAZARD DUE TO WORK WITH BUILDING / CEILING CRANE / LIFTING GEAR / LIFTING ACCESSORIES AND WORK WITH SUSPENDED LOADS	Defective lifting gear (slings, shackles, hooks, chains) - deformation, wear, missing labels Expired or missing inspection tags / certifications Incorrect rigging leading to overloading or unstable load Working under suspended loads Swinging load; unclear signals between rigger and crane operator Untrained / unqualified personnel performing lifting tasks	P.A	Dropped load injuries, fatality, equipment damage	4	3	12	M	1- Lifting gear / load rigging shall be visually inspected before use; check inspection tags for validity; mark defective lifting gear and block for further use; notify supervisor of defective lifting gear / slings. 2- Slings and shackles capable of supporting 1.25 times the weight of the load to be lifted must be used, and the slings and shackles must have label certificates (valid third-party certification, color coded). Worn or damaged equipment without labels should not be used. 3- Slings must be attached at the appropriate angle; protective pads must be used at points where they come into contact with sharp surfaces; equipment must be checked for deformation before each operation. 4- Use all load attachment rigging correctly; use provided devices; use edge protection to prevent damage to lifting gear due to sharp edges. 5- Do not work under suspended loads; maintain distance from load; keep transport routes free. 6- Do a walkdown of the lift path before crane work and warn personnel working in the hazardous area; plan all crane activities in advance and prepare the set-down area. 7- Ensure clear and precise signals between rigger and crane operator; never turn your back on the load. 8- All personnel shall clear the hazardous area if the load is to be lifted above work areas; keep personnel who are not required out of the hazardous area. 9- Always use the correct device for lifting operations; consult your supervisor if you are uncertain; use only appropriately trained and qualified personnel. 10- Wear appropriate personal protective equipment (high-visibility safety vest if necessary); Helmet EN 397, work shoes EN 345 S3, gloves EN 388. 11- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
21	HAZARD DUE TO ENTERING / EXITING CONFINED SPACE	Fall into the confined space / unguarded access openings Blocked or obstructed emergency exits Oily, unstable or slippery access surfaces Use of access points not provided for this purpose Poor lighting at entry / exit points	P	Fall injuries, entrapment, fatality	4	3	12	M	1- When accessing scaffolding / confined space, use only the access point provided for this purpose. 2- Clearly define access and exit points; keep emergency exits clear at all times. 3- Keep the work area free from oil; provide adequate lighting at entry / exit points. 4- Barricade off the work area if possible; hard barriers and warning signs at the entrance. 5- A CS attendant / watchman to be available at entry and exit area; proper housekeeping. 6- Minimum PPE as per activity (Helmet EN 397, work shoes EN 345 S3, gloves EN 388). 7- Emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
22	HAZARD DUE TO UNRESTRICTED AREA	Unauthorized persons entering the hazardous area Uncontrolled entry / exit during operations Absence of barricades and warning signs Presence of untrained persons in the work area	P.A	Personal injury, property loss, security incidents	3	3	9	M	1- Barricade off the work area if possible (hard barricading); keep personnel who are not required out of the hazardous area. 2- Use warning signs and safety tape; mark the area before the start of work. 3- Use only appropriately trained and qualified personnel in the work area. 4- Security to control the access points; all personnel to wear valid ID cards. 5- Inform subcontractors of the risks in shared areas. 6- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	3	1	3	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
23	WORKING WITH MANLIFTS	Manlift overturning on unstable / slippery ground Use by unauthorized / untrained operators Exceeding the basket capacity (SWL) Climbing on / standing on the guardrails Modified or deactivated limit switches Contact with exposed power cables and electrical trays Machine left unsecured at the end of the shift	P.A	Fall from height, electrocution, crush injury, fatality	4	3	12	M	1- Manlifts should only be used by experienced and authorized / trained operators; a trained person familiar with the ground rescue plan must be present while working at height on the manlift. 2- Visual checks must be performed by the user before each operation; periodic inspection reports to be available. 3- Before starting work, the ground must be checked to ensure it is stable and level; work should absolutely not be done on slippery surfaces. 4- When changing positions, there should be no obstacles, gaps, cables, or hoses along the route that could cause tipping; the scissors must be in the closed position. 5- Wheel locks and support legs must be used to prevent the manlift from moving during operation. 6- The capacity of the basket (Safe Working Load) must not be exceeded; it should only be used to lift workers and hand tools; maximum load information must be clearly visible on the equipment. 7- Limit switches must not be modified or deactivated; do not climb on or stand on the guardrails of the manlift. 8- At the end of the shift, the machines must be locked and placed on a solid and level surface where there are no obstacles or movement; the platform must be lowered, the ignition key removed and kept in a place inaccessible to unauthorized persons, and chocks must be placed under the wheels. 9- If welding is to be performed during operation, the welding frame must not come into contact with the machine; external grounding of the body must be provided. 10- When working on the manlift, attention must be paid to exposed power cables and electrical trays, and a safe working distance must be maintained. 11- PPE: Helmet EN 397 with chin strap, full body harness EN 361 with double lanyard, work shoes EN 345 S3. 12- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
24	HAZARD DUE TO INSUFFICIENT KNOWLEDGE IN FIRE EVACUATION	Unclear / unknown evacuation routes Personnel not trained in emergency response Emergency procedure not communicated to the workforce Muster points not identified or explained Lack of alarm awareness and response	P	Delayed evacuation, injuries, fatality	4	3	12	M	1- Necessary instructions will be communicated to the staff in accordance with the Customer Emergency procedure. 2- Muster points to be identified and explained to the workforce during the TBT. 3- An emergency action plan to be organized and communicated to all employees; evacuation routes to be clear, marked and unobstructed. 4- All crew must be familiar with the emergency procedure and alarm signals; in case of alarm triggered, immediately evacuate the plant headed to the muster point. 5- Regular fire / evacuation drills to be conducted; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
25	HAZARD DUE TO FIRE OR EXPLOSION	Mishandling of fuel and oxidizing gas cylinders Fire loads / combustible waste accumulation in the work area Absence of fire watch and extinguishing agents Smoking in the oil storage area Defective or uninspected equipment Insufficient distance between fuel and oxidation gas cylinders	P.A. E	Fire, explosion, burns, serious injury, death, asset damage	4	3	12	M	1- Only use tools which are suitable or intended for the activity; visual inspection before use. 2- Remove hoses from combustible gases and oxygen or disconnect from the supply point when work is interrupted. 3- Maintain a distance of at least 3 meters between fuel gases and oxidation gases or provide for a fire-resistant partition. 4- ATTENTION: Liquid gas cylinders (propane or butane) shall be stored at least 3 meters from ALL other types of gas cylinders. 5- NO SMOKING in the oil storage area. 6- Remove fire loads, assign fire watches, and provide for extinguishing agents; fire extinguishers to be available at all times and operated by a trained person. 7- Safety instructions in the instruction manual shall be observed; wear appropriate PPE. 8- Good housekeeping to avoid accumulation of combustible waste. 9- Conduct TBT; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L
26	HAZARD DUE TO FALLING LOADS	Dropped load from crane / lifting equipment Load not suspended in accordance with drawing specifications Personnel present under the suspended load Hand tools and objects not secured while working at height Unsecured materials and unmarked hazardous area	P.A	Struck-by injuries, fatality, property damage	4	4	16	H	1- Barricade off the work area; keep personnel who are not required out of the hazardous area. 2- Suspend the load in accordance with drawing specifications; use the correct rigging and tag lines; a competent rigger / slinger to attach the load. 3- NEVER work under a suspended load; no personnel under the load under any circumstances. 4- When working at height, all hand tools should be kept in the keeper (tool pouches / lanyards). 5- Lifting gear to be visually inspected before use; defective gear to be withdrawn from service and quarantined. 6- Slings to be suitable for the load weight and in safe condition without deformation. 7- Minimum PPE: Helmet EN 397 with chin strap, gloves EN 388, work shoes EN 345 S3. 8- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
27	WASTE MANAGEMENT	Unsorted waste (metal, cable, wood, paper, hydrocarbons) Improper disposal of hazardous waste Waste containers without lids Waste accumulation attracting wild animals Blocked routes and work areas by waste debris	E	Environmental damage, regulatory non-compliance, property loss	3	3	9	M	1- Waste must be sorted according to waste group (metal, cable, wood, paper, etc.) and waste class and disposed of in the designated waste area. 2- Waste containers with proper lids; timely evacuation of waste. 3- Hazardous waste segregation; waste disposal contract must be available; MOTAT records to be maintained. 4- Hazardous waste to be handled in accordance with the MSDS and legal / local rules by a licensed disposal contractor. 5- Proper housekeeping - waste to be removed from work areas to avoid obstruction to movement and attraction of wild animals. 6- Emergency procedure to follow.	3	1	3	L
28	HAZARD DUE TO POOR ILLUMINATION	Insufficient lighting in work areas Night / low-light work without temporary lighting Defective or untested temporary lighting Shadows and blind spots in the work area	P	Trips, falls, cuts, struck-by injuries	3	3	9	M	1- Illumination must be increased according to conditions; lighting that is appropriate to the situation shall be provided (temporary lighting for night / dark areas). 2- Adequate lighting to be ensured in all work areas before the start of work; defective lighting to be repaired / replaced immediately. 3- Personal head lamps to be used for night / low-light tasks. 4- Signage to mark hazards in low-light areas; housekeeping to be maintained. 5- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	3	1	3	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
29	LIFTING OPERATION	Overloading the crane beyond its tonnage / S.W.L. Defective lifting accessories (deformed slings, missing certification) Crane not periodically inspected; uncertified operator Soft / uneven ground under outriggers Adverse weather (high wind) during lifting Personnel underneath the material Blind lifts without radio communication Sharp edges of the load damaging the slings No distance / communication provided between load and operators	P.A	Crane overturn, dropped load, injury / fatality, asset damage	4	3	12	M	1- The weight of the material to be lifted must be known before lifting, and the crane must be selected according to the weight of the material; loads must not exceed the S.W.L. of the crane or the slings. 2- The lifting accessories to be used must have appropriate capacities and must be in a safe condition without deformation, with valid third-party certification and color code. 3- Unauthorized personnel should not be present in the work area; the lifting area must be marked, barricaded and a warning sign must be placed. 4- The crane must undergo periodic inspection, the operator must have a valid certificate, training and experience (TPI certified) and it must be checked; the daily crane inspection sheet to be completed. 5- The crane's legs must be extended, the ground checked and ensured to be level, and the crane's outriggers and pads must be used to adapt to the ground; ensure the crane setting by trial lift; soil condition to be assessed with strength calculation if required. 6- Weather risks must be assessed - in case of winds exceeding 50 km/h (lifting plan wind limit 9.5 m/s), all operations are suspended. 7- The necessary distance between the load and the operators must be provided and the personnel must be informed; the operation of the signal warning lights must be checked. 8- Absolutely no one should go underneath the material; if necessary, a rope / cord (tag line) should be used for guidance. 9- When securing the material, if the material has sharp edges, cardboard or similar material should be placed under the material to prevent damage to the slings; no load exceeding the tonnage of the slings should be attached. 10- The lifting operation must be controlled by the signalman (banksman); if visual contact is not possible, communication between the crane operator and the signalman should be ensured by providing them with a radio. 11- All personnel performing the operation must wear all necessary appropriate PPE (Helmet EN 397 with chin strap, high-visibility vest, work shoes EN 345 S3, gloves EN 388). 12- When bringing the panels into the control building, they must be lowered due to the door height; this operation must be done with the help of a crane; the slings used to lift the panels must be suitable for the weight of the panels, and deformed slings must absolutely not be used. 13- Ensure the approved lifting plan, PTW and other relevant documents are available for the lifting activity; continuous supervision by the lifting supervisor. 14- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
30	CABLE PULLING	Cable reel falling during unloading from the truck Uncontrolled crane operation during reel handling Injury from the hard tip of the fiber guide when driven through pipes Manhole hazards (confined atmosphere, sewage, rainwater accumulation) Defective cable pulling device / reel stands (no periodic inspection) Working near energized MV cables in the channel	P.A	Personal injury, cable damage, electrocution, asphyxiation	4	3	12	M	1- When removing the cable reel from the truck, a sling appropriate for the weight of the material must be used. 2- A crane operator must be used, the crane's lifting apparatus must be checked, the slinger-signaler must direct the operation, the crane's outriggers must be sufficiently spread apart, and pads must be placed under the outriggers to ensure stability on the ground. 3- No one should be in the area of operation; no one should enter under the material; a guide rope must be used, the necessary warnings must be given, and a lookout must be present. 4- The cable reel must be set up on a level surface using a cable stand suitable for the tonnage of the cable reel; there must be a sufficient number of personnel at the head of the cable reel to turn the reel. 5- No one should stand in front of the cable reel when the fiber guide is being pulled through the laid pipes, as the end is hard. 6- The cable pulling device must be anchored to the ground with dowels; the area must be enclosed with railings and barriers. 7- Cable reels should be fixed inside the manholes, and the cable should be pulled through them; manholes should be cordoned off with tape and covered at the end of the shift. 8- The number of personnel for cable pulling should be appropriate for the operation, and communication should be via telephone / radio; cable cutters should be used for cable cutting. 9- All ladders, platforms, or scaffolding used for cable pulling must be inspected by a competent person, certified with a green card, and comply with standards. 10- The cable pulling device and reel transport stands must have periodic inspection reports issued by competent institutions within the last year. 11- Where energized MV cables are located, protection must be worn and insulating mats must be laid in the channel; use rubber sheets to protect the cable at the edges of the cable ducts and proper cable rollers. 12- Minimum PPE (Helmet EN 397, work gloves EN 388, safety glasses EN 166, reflective vest, work shoes EN 345 S3); boots should be worn due to the accumulation of rainwater and sewage water inside the manhole. 13- Conduct TBT; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L
31	HAZARD DUE TO UNINTENTIONAL ACTIVATION / INCORRECT OPERATION OF SYSTEMS / COMPONENTS	Unexpected start-up / activation of equipment or systems Incorrect operation of systems / components during testing and handover Work continuing after unexpected energization Lack of communication with participating persons before activation	P.A	Injury, fatality, equipment damage	4	3	12	M	1- Permit to work has been issued; NOTE: permit to work shall be prepared and issued only by a person tested and authorized by the customer. 2- Ensure communication with all participating persons before activation and operation. 3- Deactivate pumps before checking oil strainers; LOTO to be applied for all energy isolation. 4- Ensure electrical isolation is carried out by authorized and competent personnel; isolation to be tested and verified using approved testing devices. 5- Ensure isolation details are communicated to all crew during TBT. 6- Ensure no re-energization until work completion and proper handover. 7- Only authorized and competent personnel to operate systems / components. 8- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
32	HAZARD DUE TO HANDLING OF INSULATING MATERIAL	Inhalation of dust / fibers from insulating material Skin and eye irritation Working at height during installation without fall protection Unauthorized personnel entering the work area Dust generation affecting the environment	P.E	Respiratory illness, skin / eye irritation, fall injuries, environmental impact	4	3	12	M	1- Keep personnel who are not required out of the work area; barricade the work area. 2- Wear protective clothing and respirator masks (FFP3 dust mask) in accordance with the material MSDS. 3- Use fall protection equipment (full body harness EN 361) when necessary / working at height. 4- Handle the insulating material in accordance with the MSDS instructions; minimize dust generation and use local exhaust ventilation where necessary. 5- MSDS to be available at the work area and explained to the employees during TBT. 6- Minimum PPE: Helmet EN 397, FFP3 dust mask, gloves EN 388, protective clothing, work shoes EN 345 S3. 7- Emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L
33	HAZARD DUE TO UNAUTHORIZED PERSONS / OTHER SUBCONTRACTORS IN WORK AREA	Unauthorized access to the work area Uncoordinated simultaneous work of other subcontractors Unmarked work areas and missing warning signs Lack of risk awareness of other crews operating nearby	P.A	Personal injury, property loss, security incidents	4	3	12	M	1- Mark areas and barricade off if necessary or set up warning signs; hard barricading in high-risk zones. 2- Inform subcontractors of the risks in the work area; coordinate activities in shared areas before start of work. 3- Control of access points by security; all personnel to wear valid ID cards. 4- Inform other crews of on-going operations via TBT and signage. 5- Minimum PPE in controlled areas; conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
34	ASSEMBLY OF PRIMARY EQUIPMENT - Crane operations, use of slings and shackles	Crane operations during primary equipment assembly Defective / uncertified slings and shackles (missing labels) Incorrect sling angle of attachment Sharp surfaces cutting / damaging slings Pylon climbing without double lanyard / anchor verification Personnel walking on beams during installation Unsecured anchor points for lanyards	P.A	Dropped load, injury, fatality, asset damage	4	3	12	M	1- Lifting accessories must be stored in a way that prevents damage or deterioration; ropes, slings, and chains must be replaced when deformation occurs and their certifications must be recorded. 2- Slings and shackles capable of supporting 1.25 times the weight of the load to be lifted must be used, and the slings and shackles must have label certificates. Worn or damaged equipment without labels should not be used. 3- Slings must be attached at the appropriate angle; protective pads must be used at points where they come into contact with sharp surfaces; equipment must be checked for deformation before each operation. 4- Personnel must use double lanyards (full body harness EN 361) when climbing pylons to install beams and remove slings; it must be absolutely ensured beforehand that the areas where the lanyards will be attached are securely created. 5- Long slings should be used to ensure that the slings stay close to the pylon during beam installation, preventing personnel from walking on the beam. If short slings are used, personnel will attach to the yo-yo, which is fixed to the crane hook, to remove the slings. 6- Ascending the pylon should be done using a vertical lifeline and/or yo-yo; walking on the equipment should be done using a horizontal lifeline. 7- Only trained and competent rigger / banksman; radio communication for blind areas; continuous supervision. 8- Minimum PPE: Helmet EN 397 with chin strap, EN 361 harness with double lanyard, work shoes EN 345 S3, gloves EN 388. 9- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
35	CURRENT TRANSFORMER , VOLTAGE TRANSFORMER, SURGE ARRESTER - Installation of Supports	Improper preparation of the ground assembly work area (slip / trip / fall factors) Sling damage at the support attachment points (sharp edges) Work at height during support installation without harness Retrieval of the sling from the top after assembly Uncontrolled access of personnel to the work area	P.A	Personal injury, property damage	4	3	12	M	1- Before starting ground assembly work, the work area must be prepared and organized; factors that could cause slipping, tripping, and falling must be eliminated. 2- Sling attachment at the supports should be done at the top, and a protective pad or similar piece should be used under the sling to prevent damage. 3- Personnel must work with a safety harness (EN 361) where work at height is involved. 4- After the support assembly is complete, the sling should be retrieved from the top using an appropriate type of ladder (per the portable ladder usage requirements of this assessment). 5- Work permit to be obtained before work; strict compliance to PTW requirements. 6- Minimum PPE: Helmet EN 397, work shoes EN 345 S3, gloves EN 388, safety glasses EN 166. 7- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
36	WILD ANIMAL BITES / EXPOSURE TO DISEASE-CARRYING VECTORS	Venomous snakes, scorpions and wild dogs in the study area Mosquito-borne and vector-borne diseases Inadequate routine vaccination of personnel Uncontrolled food waste / still water attracting animals and vectors Drivers exposed to wild / domestic animals on the road	P	Bites, injuries, infection, death	2	4	8	M	1- In mild injuries, the wound should be washed with soap and cold water for 5 minutes and covered with a clean cloth. In case of serious injury and bleeding, the bleeding should be stopped by applying pressure to the wound with a clean cloth and support should be received immediately from 112. 2- The venomous animal situation for wild animals belonging to the study area should be learned from health centers; the worker should be transferred to the hospital in possible snake bites and it should be ensured that snake serum is intervened against snake bites. 3- Comprehensive information on common local vectors and diseases should be collected; if necessary, special expertise should be acquired at the first stage. 4- Personnel to be trained in first aid; first aid kit to be available; certified first aider at site. 5- Emergency Iraqi numbers (112) must be available; all personnel familiar with the emergency procedure. 6- Personal behavioral training should be provided (e.g., food disposal, clothing, fly repellents); mechanical vector control measures should be taken (e.g., mosquito screens, mosquito nets); reproduction control should be done (e.g., disposal of still water); chemical or biological vectors should be controlled (applied by experts). 7- If there is a deficiency in routine vaccination, it should be completed; if possible and necessary, additional immunization should be done. 8- Inspect personal clothing and PPE before use for presence of scorpions; inspect the working area for presence of venomous animals and insects. 9- While driving, be careful about the possibility of wild or domestic animals on the road.	1	4	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
37	CABLE LAYING	Cable reel unloading from the truck (dropped load, sling failure) Uncontrolled cable reel turning (personnel at the head of the reel) Injury from the hard tip of the fiber guide in laid pipes Manhole entry (confined space, sewage, rainwater) Cable damage at duct edges / supports Working near energized MV cables in the channel	P.A	Personal injury, electrocution, asphyxiation, cable damage	4	3	12	M	1- When removing the cable reel from the truck, a sling suitable for the tonnage of the material must be used. 2- The crane must be operated by a qualified operator; the lifting apparatus of the crane must be checked; the sling operator / signaler must direct the operation; the crane's outriggers must have sufficient clearance and pads must be placed under the legs for adaptation to the ground. 3- No one should be in the operation area; absolutely no one should go under the material; a guide rope must be used, necessary warnings must be given, and a lookout must be present. 4- The cable reel must be set up on a flat surface using a cable stand suitable for its tonnage; there must be enough personnel at the head of the cable reel to turn the reel. 5- No one should stand in front of the fiber guide when it is being driven through the laid pipes because its tip is hard. 6- The cable pulling device must be fixed to the ground with dowels; the area must be enclosed with railings and barriers. 7- The cable pulling device and reel carrying stands must have periodic inspection reports issued by competent institutions within the last year. 8- Cable reels should be fixed inside the manholes, and the cable should be pulled through them; manholes should be cordoned off with tape and covered at the end of the shift. 9- The number of personnel for cable pulling should be appropriate for the operation, and communication should be via telephone; cable cutters should be used for cable cutting. 10- Where energized MV cables are located, protection must be worn and insulating mats must be laid in the channel. 11- Personal protective equipment should be worn and used (Helmet EN 397, work gloves EN 388, safety glasses EN 166, reflective vest, work shoes EN 345 S3); boots should be worn due to the accumulation of rainwater and sewage water inside the manhole. 12- Conduct TBT; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L
38	SWITCHYARD CIRCUIT BREAKER INSTALLATION WORK	Lifting of heavy circuit breakers by crane Unprepared work area (hazards not identified by site survey) Work at height without fall protection Simultaneous (top-bottom) work during mechanical assembly Material tipping / tilting during lowering from the upper section Energized points within the safety distance	P.A	Serious injury, fatality, asset damage	4	3	12	M	1- Before work begins, site officials must conduct a site survey to identify hazards and risks; work should not commence without taking the necessary precautions. 2- Before mechanical assembly work begins, appropriate work permits must be obtained and work permit requirements must be met (strict compliance to PTW). 3- Work at height requirements specified in the "Loading the primary material to the site" work step must be ensured (full body harness EN 361 with 100% tie-off). 4- Crane use and safety precautions specified in the "Mechanical assembly work" work step must be ensured (certified crane, signalman directing the operation, barricading, no personnel under the load). 5- Materials will be lifted using 2 slings and by lowering them from the upper section; precautions will be taken against material tipping, tilting, etc. (tag lines, competent rigger). 6- The power must be cut off at points within the safety distance, and the 5 safety rules must be applied (Isolate, LOTO, verify, ground and short circuit, maintain isolation). 7- Minimum PPE: Helmet EN 397 with chin strap, EN 361 harness, work shoes EN 345 S3, gloves EN 388. 8- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Con sequences)	L	S	R	Rat e	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rat e
39	LANDSCAPING AND EXCAVATION WORKS	Movement of trucks / construction equipment (excavator, JCB, loader) - blind spots, collision, overturning Striking or damaging underground utilities / facilities Untrained or uncertified drivers Excavated material spilling from trucks Personnel present near the truck during loading Water accumulation inside the excavation	P.A. E	Personal injury, fatality, damage to underground utilities, environmental impact	4	3	12	M	1- The licenses, maintenance records, and other legal documents for trucks and construction equipment (JCB, Loader, Excavator, etc.) to be used on site must be provided; certified equipment / machines with approved valid certification. 2- Drivers' personal documents must be submitted, and they must hold a valid driving license as per law and receive on-the-job training; certified driver / operator only. 3- An additional observer / flagman with reflective vest must be present in the maneuvering and working area of trucks and construction equipment; audible reverse alarm to be installed. 4- Workers must be prevented from entering the truck's blind spot, and training must be provided to workers on this subject. 5- No one should be near the truck when loading excavated material onto it. 6- Excavated material must be transported to the area indicated by the customer; equipment must be available on trucks to prevent excavated material from spilling. 7- Ensure all live wires / cables must be identified prior to start of excavation in the area; obtain excavation certificate and scanning report; always keep the certificate attached with the PTW. 8- Provide mandatory side-wall protection (shoring, sloping, or benching) for all excavations exceeding 1.2 meters in depth; spoil soil, excavated materials, and heavy equipment must be kept at a minimum distance of 1.5 meters from the edge; at least two accesses with proper slope / ladder to be kept without obstructions. 9- If water accumulation is present, immediately provide a water pump (de-watering). 10- Erect barricade and warning signs properly (hard barricading); maintain housekeeping standard at all times. 11- Minimum PPE: Helmet EN 397, work shoes EN 345 S3, safety glasses EN 166, gloves EN 388, high-visibility vest. 12- Conduct TBT; emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	4	1	4	L
40	GENERAL FIELD ACTIVITIES - WORKING NEAR ENERGIZED POINTS	Approaching energized equipment beyond the safe distance Work within the safety distance without energy shutdown Employees not trained on energy points and safe approach distances Raising materials above energized points Energized areas not restricted / barricaded	P.A	Electric shock, electrocution, arc flash, burns, fatality	4	3	12	M	1- Energized areas must be restricted using appropriate barriers, tape, and warning signs; clear identification of live equipment. 2- Work must be performed using equipment that will not approach the safety distance (excavators, etc.); the safety distance must be calculated prior to work. 3- Energy shutdown must be requested for work within the safety distance, and the 5 safety rules must be applied (Isolate, LOTO, verify, ground and short circuit, maintain isolation). 4- Employees must be trained on energy points and safe approach distances; verify line status before work; maintain safe clearance. 5- Employees must not raise materials they are holding upwards near energized points. 6- Stop work immediately if safe distances cannot be maintained. 7- Minimum PPE: insulated gloves where required, Helmet EN 397, face shield / safety glasses EN 166, work shoes EN 345 S3. 8- Conduct TBT; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L

N o.	Pathway of Activity	Hazard Description	Target	Effects (Consequences)	L	S	R	Rate	CONTROLS & RECOVERY MEASURES REQUIRED	L	S	R	Rate
41	EMERGENCY SITUATION	Unorganized emergency response plan Emergency response teams not identified or trained Contact information of emergency teams not posted / unknown Inadequate first aid and emergency kit contents Workers not identifiable (no ID card) during emergency	P.A	Delayed rescue, aggravated injuries, fatality	4	3	12	M	1- An emergency action plan should be organized and communicated to all employees. 2- Teams that will respond to emergencies (first aid team, fire team, search and rescue team, evacuation team, etc.) should be identified and trained, and the training provided should be recorded. 3- The list of names and contact information for emergency response teams must be posted in a visible location in the work area and mobilization area (112, nearest hospital, response times). 4- The contents of the first aid and emergency kits must be determined by the occupational safety specialist and technical authorities, procured, and kept on site. 5- Every worker on the construction site should be given an ID card; the back of the ID card should contain the project emergency numbers, and the front should contain the employee's identity information and the training they have received. 6- Certified first aider must be available at site at all times; emergency procedure to follow. NO HSE - NO WORK.	4	1	4	L
42	WORKING WITH CHEMICALS	Exposure to chemical fumes, dust or fog (ethyl alcohol, acid, base, epoxy, etc.) Spillage / leakage of chemicals Improper storage and minimum-level control of chemicals Incompatible chemicals stored together Inadequate PPE / ignoring the MSDS requirements Improper disposal of chemical waste	P.A. E	Chemical burns, respiratory illness, poisoning, environmental damage, fire	4	3	12	M	1- Chemicals that are harmless (least hazardous) to health should be chosen where possible. 2- It should be used taking into account the MSDS (Material Safety Data Sheet); the MSDS must be available at the work area and explained to the employees during TBT. 3- Chemical wastes should be disposed of according to legal and local rules; hazardous waste segregation with a licensed disposal contractor. 4- Spilled chemicals should be cleaned immediately with absorbent and cleaning materials; absorbent (spill) kits must be available at the work area. 5- If hydraulic fluid is used, it should be used in accordance with the MSDS form. 6- It should be stored at a minimum level, in designated areas with proper containers and labeling. 7- In order not to be exposed to chemical fumes, dust or fog, suction of the air in the environment, cleaning of the air, or protective masks (FFP3) should be used; local exhaust ventilation where necessary. 8- Personal Protective Equipment specified in the MSDS of the chemicals should be used (chemical-resistant gloves, goggles, protective clothing). 9- If chemicals spill or leak into the environment, they should be cleaned with absorbent kits and reported to the EHS Manager immediately. 10- Emergency procedure to follow; certified first aider at site. NO HSE - NO WORK.	2	3	6	L

S \ L	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15
4	4	8	12	16	20
5	5	10	15	20	25

SEVERITY (S)	1 - No Injury 2 - Minor Injury 3 - Lost Time Injury 4 - Serious Injury 5 - Death or Permanent Disability
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LIKELIHOOD (L)	1 - Very Unlikely 2 - Unlikely 3 - Likely 4 - Very Likely 5 - Certain
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1 to 6	Low Risk (L)	Acceptable, Monitor & Review
8 to 12	Medium Risk (M)	Acceptable With Controls
> 12	High Risk (H)	Unacceptable - contact EHS department for guidance

NOTE: This revision uses the Siemens Energy 5x5 Severity/Likelihood matrix exclusively (per project requirement). The AIMial risk matrix is NOT used in this document.

Relevant PPE is selected to protect against the risks mentioned in Step 2. Under each PPE, the related risk number(s) from Step 2 are referenced to confirm.

PPE & Equipment	Standard / Type	Applicable to (Step 2 No.)
Helmet	EN 397 (chin strap for teams working at height)	All steps 1-42 (chin strap mandatory: 7, 14, 16, 23, 29, 30, 34, 35, 37, 38)
Work shoes	EN 345 S3 or S1P (sole and toe protection, anti-slip)	All steps 1-42 (boots for manhole work: 30, 37)
Safety glasses / Eye shield	EN 166 (anti-fog)	All steps 1-42 (full-coverage goggles / face visor: 11, 17, 25, 34)
Ear protection	Ear plugs / muffs	11, 17, 25, 28, 29
Safety gloves	EN 388 (task-specific)	All steps 1-42
Insulated / arc-rated gloves	Electrical grade (per task)	9, 30, 37, 40
Dust mask	FFP3	1, 11, 15, 30, 32, 37, 42
Face shield	Full coverage (hot work)	11, 17, 25
Full body safety harness	EN 361 with double lanyard (yoyo) / lifeline	7, 14, 16, 20, 23, 26, 29, 32, 34, 35, 37, 38
Reflective high-visibility clothing	High-visibility vest	All steps 1-42 (mandatory in traffic & lifting areas: 3, 10, 12, 29, 39)
Disposable overalls / work outfit	Company logo work outfit	All steps 1-42 (protective clothing per MSDS: 32, 42)
Insulating mats	Electrical mat (energized MV channels)	9, 30, 37, 40
LOTO kit	Locks, tags, hasps, voltage testing device	9, 15, 31, 40
Fire extinguisher / fire blanket	ABC / CO2 (per zone)	11, 12, 25, 29, 39
First aid kit	Per project standard	All steps 1-42

The site manager / field supervisor and the OHS specialist conduct the final review and mark the appropriate section. If 'UNSUITABLE' is indicated, this hazard assessment must be redone. Work cannot start during this time.

Role	Name / Surname	Decision	Date	Signature	Phone
Head of Project Manager	Ugur Akbulut	SUITABLE () UNSUITABLE ()			
Project Manager	Onur Ozvatan	SUITABLE () UNSUITABLE ()			
Site Manager مدير الموقع		SUITABLE () UNSUITABLE ()			
EHS Manager السلامة والصحة مدير البيئة والصحة		SUITABLE () UNSUITABLE ()			
Quality Manager	Abdurrahman Ogut	SUITABLE () UNSUITABLE ()			

Trainer / Sign:	
Training Area:	
Training Date:	

I guarantee that I will follow the rules mentioned on the precautions part of the risk assessment form, and also under negative conditions I will stop working in a safe way, then inform my project manager.

No.	Name / Surname	Signature
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